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CUSTOMER SEGMENTATION REPORT

Insights from Clustering Analysis on E-Commerce
Transaction (2012-2016)

Abstract

This report analyses ~952,000 e-commerce transactions across 47 countries, identifying five distinct customer groups through feature engineering and clustering. The insights enable sharper marketing, smarter resource allocation, and stronger customer value.

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Table of Contents

<i>Introduction</i>	2
<i>Methods</i>	3
Interpreting the Dendrogram	5
<i>Results</i>	7
<i>Conclusions</i>	10
<i>Summary</i>	11

Introduction

This project developed a **customer segmentation strategy for a global e-commerce business** operating in 47 countries and five continents. The **dataset contained nearly 952,000 transactions** from 2012 to 2016, covering demographic, geographic, and purchase-level attributes.

The goal was to transform transactional data into customer-level insights that improve targeting, retention, and profitability. Order records were aggregated into unique customer profiles, and key behavioural features such as **Age, Recency, Frequency, Customer Lifetime Value (CLV), and Average Unit Cost** were engineered to capture spending patterns and loyalty.

Unsupervised learning techniques including K-means clustering (with Elbow and Silhouette validation) and Hierarchical clustering were applied, alongside dimensionality reduction methods such as PCA and t-SNE.

The resulting segments help the business identify high-value customers, churn-risk profiles, and price-sensitive groups, enabling more effective marketing campaigns, better resource allocation, and stronger long-term customer engagement.

Methods

The methodology was designed to transition from raw transactional data to actionable customer segments.

1. **Data Preparation & Feature Engineering:** Order-level records were first aggregated to create a single profile for each unique customer. From this customer-level data, key behavioural features were engineered:

- **Recency:** Days since customer's last purchase.
- **Frequency:** Total number of transactions made by the customer.
- **Customer Lifetime Value (CLV):** Total monetary value of all purchases.
- **Average Unit Cost:** The average price of items purchased
- **Age:** The customer's age derived from the available data.

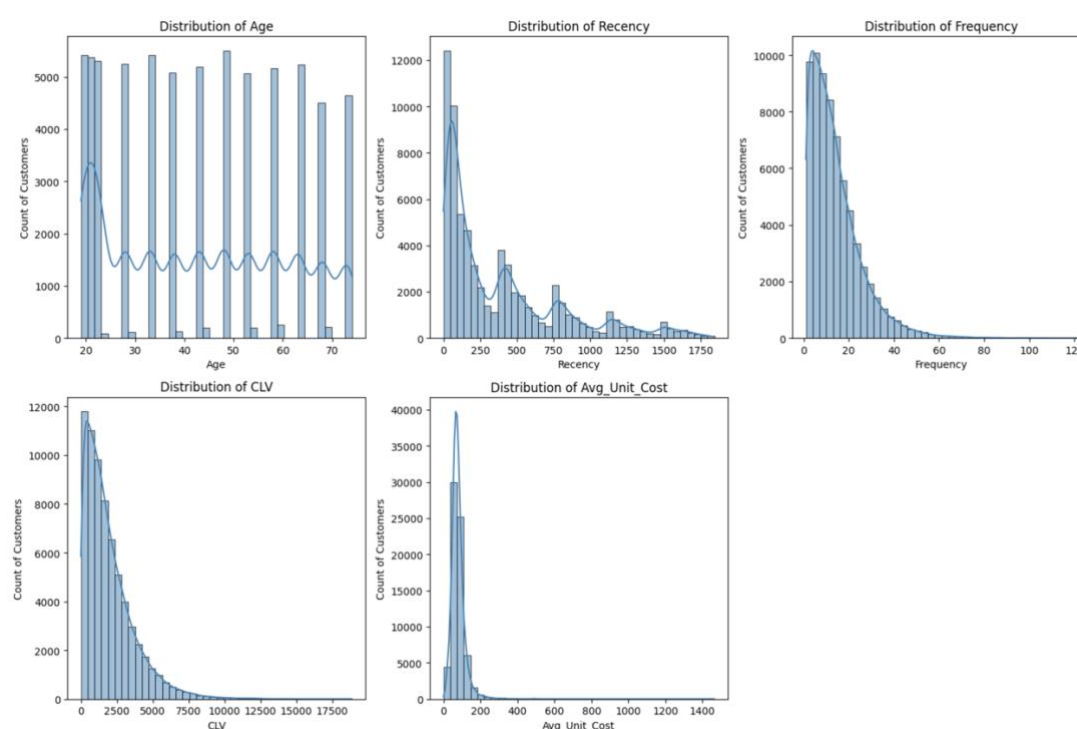


Figure 1 - Histogram of 5 Newly Engineered Features

2. **Data Scaling:** Before Modelling, these features were standardized using *StandardScaler* to ensure no single feature disproportionately influenced the clustering algorithm.

3. **Clustering Model Selection: K-means** clustering was selected as the primary algorithm. To determine the optimal number of segments (k), three validation techniques were employed:

- **Elbow Method:** This method identified a range of potential values for “ k ” by plotting the sum of squared distances and finding the “elbow” point where the rate of decrease sharpens.

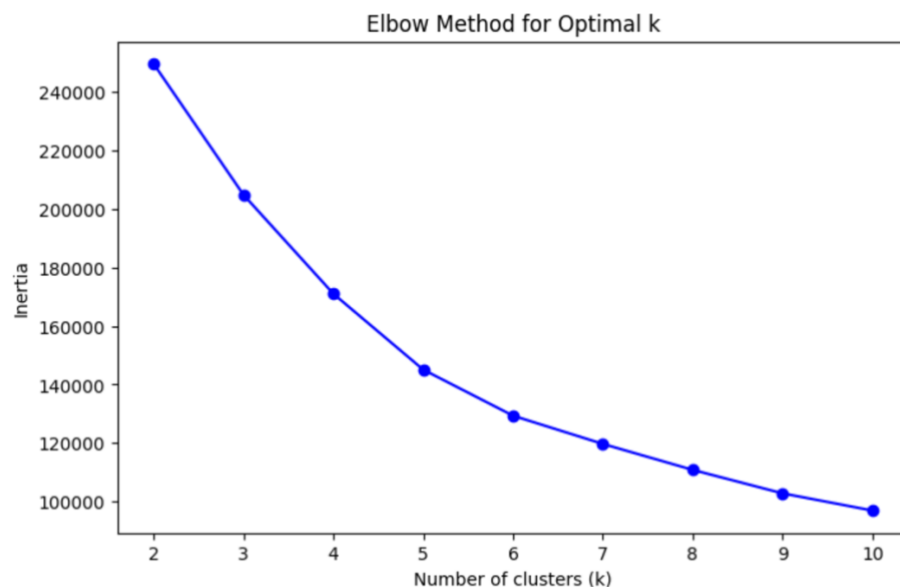


Figure 2 - Elbow Method

The Elbow curve drops sharply until $k = 4$, with only minor gains after $k = 5$. Thus, the optimal k is likely **4 or 5**.

We will validate by running the **Silhouette Score** for $k = 2$ to 7 .

- **Silhouette Method:** In this analysis, the value of “ k ” that has the **highest average score** is considered the best fit.

Clusters (k)	Silhouette Score
2	0.2622
3	0.2527
4	0.2533
5	0.2671
6	0.2526
7	0.2349

The results from the Silhouette analysis indicates that the Peak is at $k = 5$ (0.2671), which matches the results from the Elbow Flattening method earlier.

Finally, we’ll perform Hierarchical Clustering as our final validation.

- **Hierarchical Clustering:** This method builds a hierarchy of clusters by progressively merging or splitting groups based on their similarity or distance. However, for this analysis, **Hierarchical clustering was performed on a subset of 30,000 samples**, since running on all the full datasets would exceed memory limits and crash the notebook.

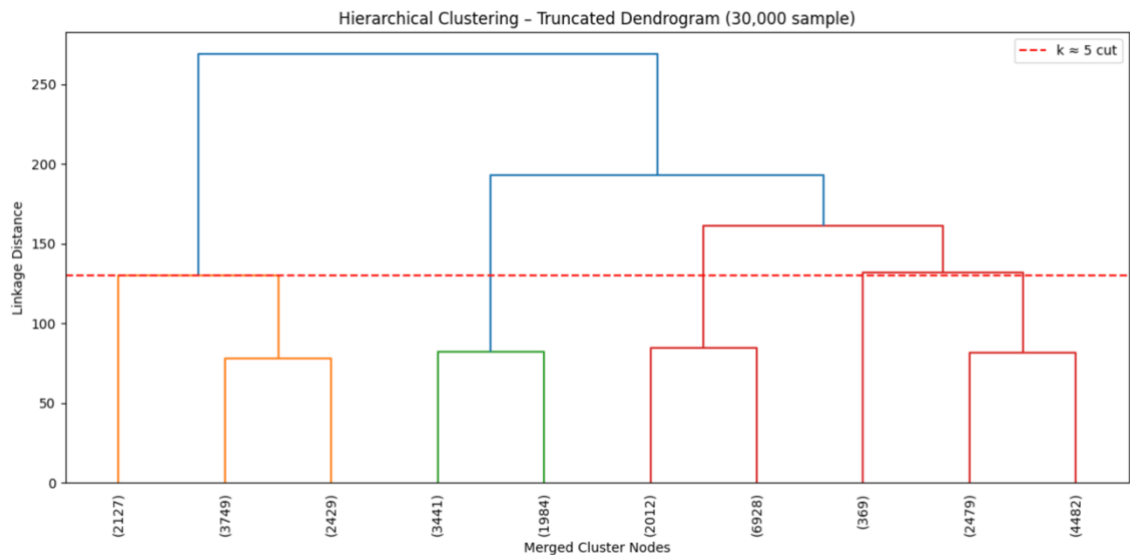


Figure 3 - Dendrogram of 30,000 subsets of dataset

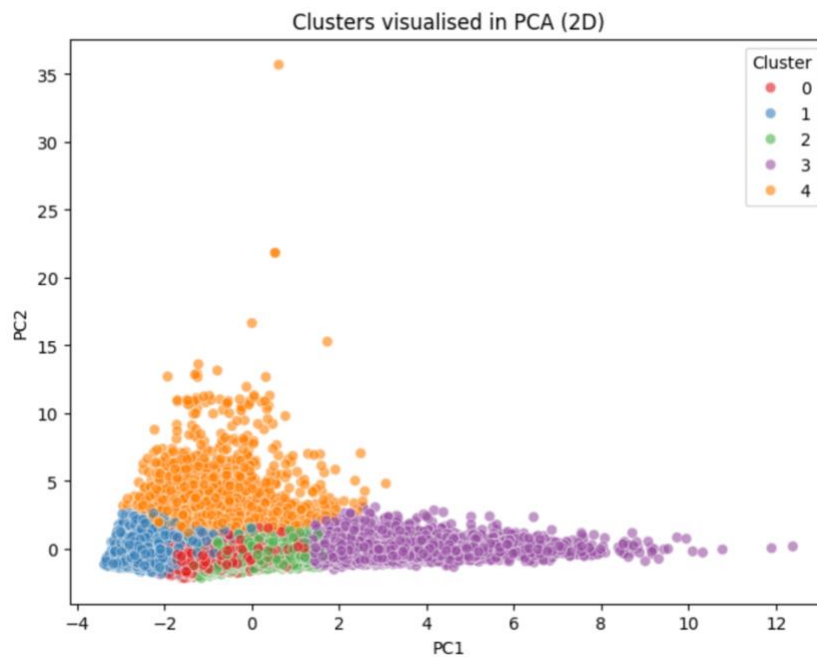
Interpreting the Dendrogram

The dendrogram alone **doesn't strongly justify $k = 5$** — it appears somewhat arbitrary. **A clearer structural break occurs around $k = 3$** , where the hierarchy shows a sharper separation before the major linkage jump.

- This suggests the data exhibits a **multi-scale cluster structure**: $K = 3$ captures broader groupings, while $K = 5$ offers finer segmentation.
- **For consistency with the Elbow and Silhouette analyses, we proceed with $k = 5$ as the chosen solution.**

4. **Dimensionality Reduction:** To visualize the resulting high-dimensional segments, **Principal Component Analysis (PCA)** and **t-SNE** were utilized to reduce the features into a two-dimensional space.

- **Principal Component Analysis (PCA):** We used PCA technique to project the 5-dimensional scaled feature space into 2D for quick visual validation of cluster separation.



- **t-Distributed Stochastic Neighbour Embedding (t-SNE):** This is a non-linear dimensionality reduction technique which we've used on a subsample of 10,000 rows to visualise multiple **perplexity** ranges to see which provided the clearest separation amongst all segments.

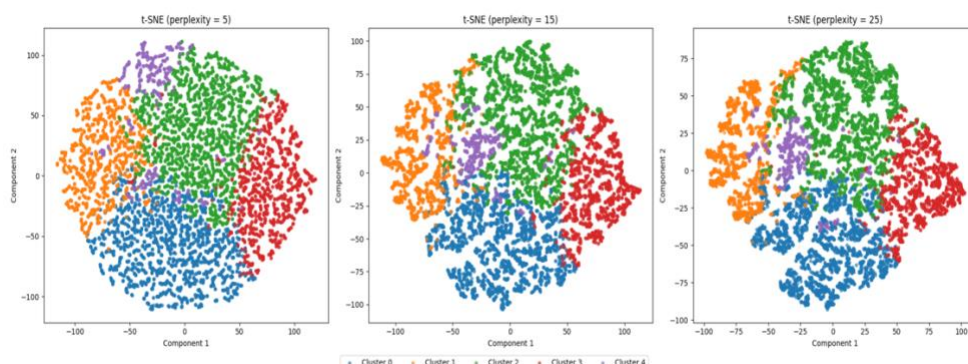


Figure 4 - t-SNE across Perplexity range 5, 15, 25

Results

The K-means clustering analysis successfully segmented the customer base into five distinct, behaviourally driven groups, each with a unique profile:

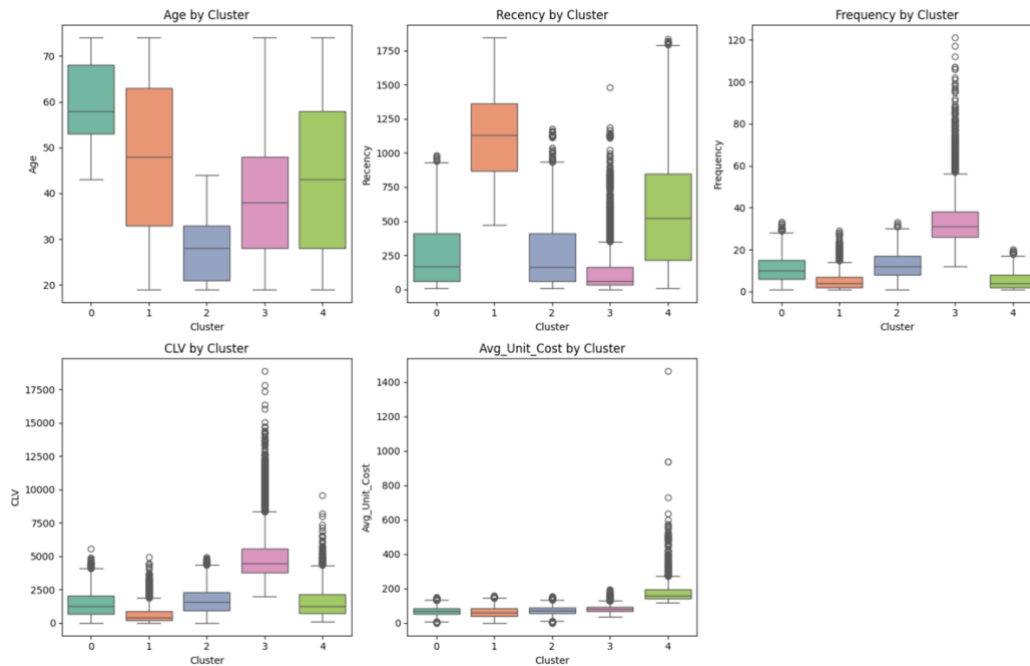


Figure 5 - Feature distribution across Five Clusters

Feature distributions across the five K-Means clusters (k = 5):

Age by Cluster

- Clusters differ noticeably by age level, with Cluster 2 representing the youngest group and Cluster 0 the oldest.

Recency by Cluster

- Cluster 3 has the lowest recency → most recent purchasers.
- **Recency cleanly separates active versus inactive clusters.**

Frequency by Cluster

- Cluster 3 has the highest median and widest spread → most frequent transactions.
- Clusters 0, 1, 4 display low frequency with tight distributions.
- **Cluster 3 clearly drives the most transactions overall.**

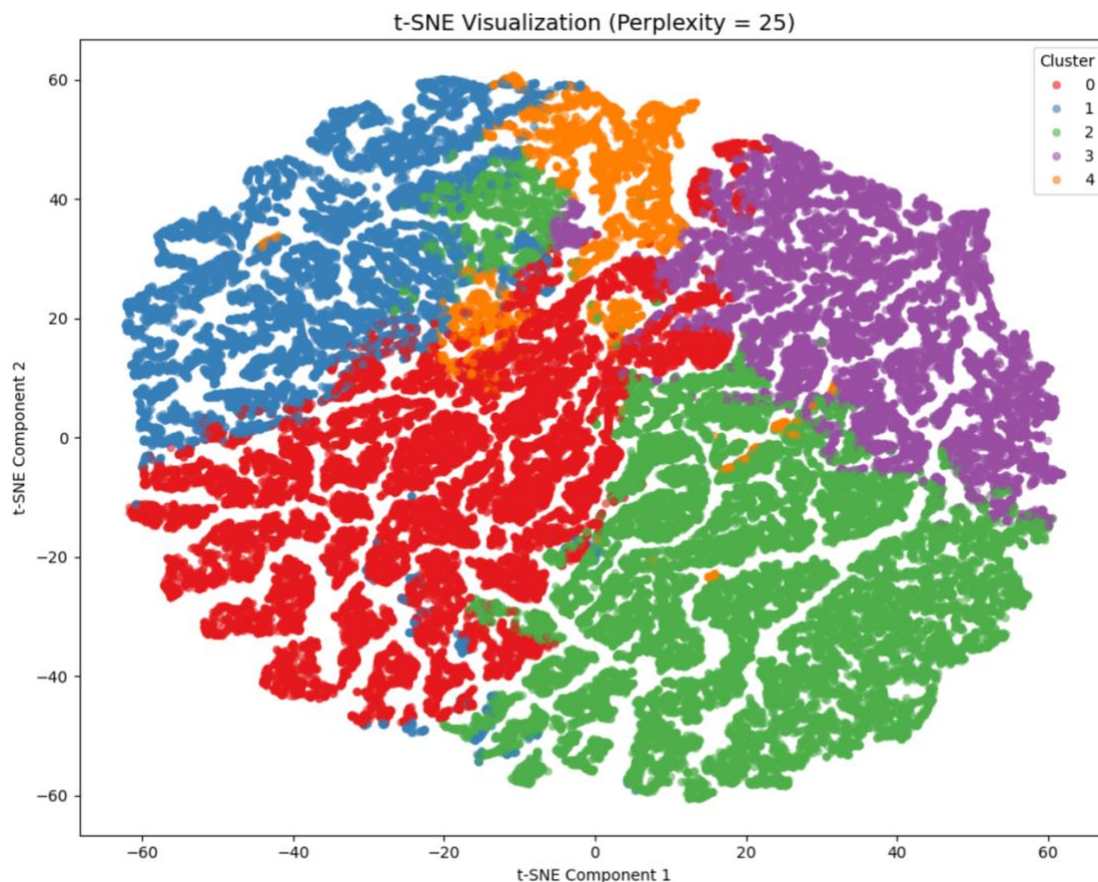
CLV by Cluster

- Cluster 3 shows the highest median CLV and many upper-end outliers → strong contribution to total value.
- Cluster 4 also shows elevated CLV with several extreme outliers.
- **Clusters 3 and 4 dominate overall lifetime value.**

Average Unit Cost by Cluster

- **Cluster 4** stands out with the highest median and large outlier range → purchases involve higher-priced items.
- Other clusters show similar, lower medians and tight distributions.
- **Average spend per unit is mostly consistent except for Cluster 4, which operates at a higher price level.**

Full t-SNE Visualization: A perplexity of 25 proved very effective for visualising the full datasets as it clearly shows the separations across all segments.



- **Cluster 0: Steady Shoppers (33.3% of base)** This group forms the reliable core of the business. They exhibit moderate recency, average frequency, and consistent, moderate spending.
- **Cluster 1: At-Risk Customers (21.6% of base)** These customers show high recency (meaning they have not purchased in a long time) and have low frequency and CLV. This segment is disengaged and at a high risk of churn.
- **Cluster 2: Budget-Conscious (22.5% of base)** This segment is defined by recent purchase activity but demonstrates minimal purchase frequency and low overall spending. This group likely consists of new customers or infrequent, price-sensitive buyers.
- **Cluster 3: Loyalists (19.8% of base)** This is a high-value segment. They have low recency (purchase often), high frequency, and a high CLV. They are highly engaged, profitable, and form a core loyal customer base
- **Cluster 4: Big Spenders (2.8% of base)** A small but critical group, these customers have low recency and low frequency, but the highest CLV and Average Unit Cost. They make large, infrequent purchases of high-value items.

Conclusions

The segmentation analysis confirms that the e-commerce customer base is not a single uniform group but is composed of five distinct segments with clear behavioural and value differences. This insight is critical for strategic planning.

Recognising these patterns allows the business to evolve from broad, "one-size-fits-all" marketing campaigns to highly targeted strategies that align directly with specific customer needs and spending behaviours. This data-driven framework provides a clear understanding of the company's core steady shoppers, high-value loyalists, budget-conscious buyers, at-risk customers, and high-value big spenders.

By tailoring marketing, retention, and engagement efforts to each specific segment, the business can significantly sharpen its marketing efficiency, allocate resources more effectively, and strengthen long-term customer relationships, ultimately driving growth and improving lifetime value.

Summary

This project successfully transformed a large transactional dataset into an actionable customer segmentation model. By aggregating order data and engineering key features like Recency, Frequency, and CLV, a K-means clustering algorithm identified five distinct customer segments. These segments profile customers as 'Steady Shoppers', 'At-Risk', 'Budget-Conscious', 'Loyalists', and 'Big Spenders'. This framework empowers the business to move beyond generic marketing and develop targeted strategies to improve retention, increase profitability, and enhance customer engagement.